Published-IC reference arm: first pilot cell (pre-freeze), 2026-10-30

Card `XP-20261024-published-ic-reference-arm`. **Status: pre-freeze pilot.** `PROTOCOL.md` is still a DRAFT and says no cell is timed before freeze, so **nothing here is a Register-E row** and nothing here supports a "vs published index calculus" claim. Labels: *measured* (a recorded value), *derived* (arithmetic shown), *projection* (derived from fewer measurements than the cell it names).

Correction to commit `cfea7e5c44`. That commit's report priced instructions at 604.6 per GAE and bounded the lab arm by a "harness noise floor". Both were wrong: `/usr/bin/time -l` on the `ecbench` parent excludes the per-run child processes in which every arm runs. This version replaces them with per-run child measurements (section 3).

1. Answer

A published-method cell ran. On Koblitz K_0 n = 9 (r = 127, m = 3, standard x-base of dimension 4), whole solve, 8 public targets, 8/8 verified `[d]G = Q`:

- msolve arm: about 1.0 x 10^12 instructions retired per target (*measured*, repeat within 0.08 %).
- lab arm (inherited F4, same session law, same targets): about 3.9 x 10^8 per target, the whole run including about 3.5 x 10^7 of process start-up (*measured*).
- Ratio per target, same unit, same host: **median 2.6 x 10^3** (range 2.2 x 10^3 to 3.1 x 10^3).
- Priced in GAE at 3265 instructions per GAE (*derived*): published arm S about 2.8 x 10^7, lab arm S about 9.5 x 10^3, matched rho 5.90. With the solver left uncharged, as `ecbench` does for this base, the two IC arms tie at 36.0.

K_0 n = 13 did **not** complete as a cell: one msolve system takes 5.70 x 10^12 instructions (544 s wall under load), and the lab arm's recorded stream needs about 36 per target. The lab arm itself costs 1.9 x 10^9 instructions per target there (*measured*), so the projected ratio is 1 x 10^5 (e_i form) or 2 x 10^4 (x-only form).

The protocol's cross-unit control, run on the lab arm at K13, **fails**: native-count pricing gives S = 631, instruction pricing gives 1.28 x 10^4 (a factor 20; the tolerance proposed in PROTOCOL section 4 is 1.5). By the protocol's own rule every arm is then priced in instructions, which is what this report does.

This says what *untuned, generic msolve F4 on the unspecialised Weil-descended system* costs. It does not say what the best published pipeline costs: no hybrid (variable specialisation) sweep, no `l` sweep, no `T` invariants, no Galbraith-Gebregiyorgis subspace, no WDSat were run (section 2). The lab's inherited-F4 solver splits on variables and is itself close to a published hybrid approach, so the ratio is as much a statement about msolve's default on these systems as about the lab. The distance to the *tuned* published arm is unmeasured.

2. Arm as run (and where it deviates from `PROTOCOL.md`)

- Instance: `icv1-f2m9-t5-81e744be` (Koblitz K_0, `KoblitzCurve::new(0, 9)`, #E = 508, r = 127, cofactor 4), base `binary-standard-subspace:dimension=4` (x in span(1, z, z^2, z^3)), m = 3; the 8 public targets of ecbench's `target_seed 202610111901` law. **n = 9 is not a**

size in PROTOCOL section 2 (the ladder starts at 13); it is used because n = 13 cannot be afforded (section 5). The msolve arm (session `ECBS1h3c8ebd974409`) shares instance, targets and query stream with the lab and rho arms of that session; their counted values are identical to the later per-run-timed sessions `B9r`/`A9r` (checked: same S per target), which are the source of the lab arm's instructions.

- Binaries (`host.json build.binary_sha256`): `cf26250a...` for `ECBS1h3c8ebd974409` and `ECBS1hdda926241d89` (built from a `git archive` of `143ea46578` plus the engine in this commit); `6cf96928...` for the per-run-timed sessions, the same plus a **measurement shim** that runs each child under `/usr/bin/time -l` (`pilot_20261030/ecbench_child_time_shim.patch`, **not committed to `runner.rs`**). The shim changes no counted value.
- Published arm = `pdp3-binary:m=3,engine=msolve-s4` (new, `src/bin/ecbench/msolve_s4.rs`): the `e_i`-symmetrised Weil-descended Semaev `S_4` of `binary_semaev_s4` (descended `S_4`, quadratic in `e_1,e_2,e_3`, plus the cubic correspondence `e_i = sigma_i(x)`), 9l - 3 = 33 unknowns at l = 4, field equations appended, solved by `/opt/homebrew/bin/msolve` 0.9.5, `-g 2 -t 1`, one process per system. Roots from the reduced basis (section 4), lifted through the framework's `lift_abscissae`, relation sign-resolved and re-added in the group, then the pipeline's rank, linear algebra and `[d]G = Q` verification. A tuning-probe form `MSOLVE_S4_FORM=x` substitutes `e_i = sigma_i(x)` and solves over the 3l x-bits only; it ran in the probe, not in a session.
- Deviations from `PROTOCOL.md`, recorded rather than silent:
  1. `S_3`-symmetrisation only; **no 2-torsion `T` invariants** (`binary_semaev_s4.rs` has none; PROTOCOL section 2′s "T-symmetrised" and section 1′s "`e_i`-symmetrised" disagree, defect D6).
  2. msolve **0.9.5** (sha256 `eb640c7a472d894a...`), not the pinned 0.6.5; no version A/A.
  3. **No tuning sweep** (`l`, `LP`, `K`, specialisation); **no Galbraith-Gebregiyorgis subspace**; **no WDSat**; m = 3 only.
  4. Pricing is **not** the protocol's Callgrind `Ir`: no Valgrind on this host. Instructions retired from `/usr/bin/time -l` (Apple M4 Pro, aarch64) stand in. Instruction counts are not comparable with the leaderboard's x86 hosts, only within this pilot.
  5. Isolation: macOS, `ecbench --cpus none`, every record is L0 (PROTOCOL asks L1). Counted metrics are unaffected; host load was 10-50, so wall times in the records are not claims and are not used.

3. Pricing and the cross-unit control

- **Instructions per run.** `ecbench` runs every arm in a child (`ecbench exec`); `/usr/bin/time -l` on the parent reports only the parent (2.8 x 10^9, about the same for rho-only and lab-only sessions). The shim wraps each child instead. Measured on this host: rho-only child about 3.5 x 10^7 per run (start-up, 55-81 GAE of work), lab child at n = 9 about 3.9 x 10^8 (3.3 x 10^8 to 4.9 x 10^8), at K13 about 1.9 x 10^9 (1.5 x 10^9 to 2.4 x 10^9). msolve's own instructions come from `/usr/bin/time -l` around each call, recorded as `solver.extra.msolve_ir`; repeat run on the same targets: total 8.5403 x 10^12 against 8.5366 x 10^12, each target within 0.08 %.
- **Instructions per GAE** (*derived*): rho-negation runs on K_0 n = 41, 4 targets, per-run child instructions less the 3.51 x 10^7 start-up over the run's GAE: 3265, 3264, 3260, 3281 (GAE 4.8 x 10^5, 7.3 x 10^5, 1.4 x 10^6, 1.4 x 10^5); n = 37 gives 3.2 x 10^3 to 3.4 x 10^3. Use **3265**. It was measured at n = 41, where an addition is dearer than at n = 9 or 13, and includes walk

overhead, so it is probably high for the small instances: S figures in instructions-priced GAE are then under-statements for both arms; the Ir-to-Ir ratio is unaffected by it.

- **Cross-unit control, lab arm, K13** (`B13r`, 8 targets, inherited F4): this tree prices the lab's word XORs through the pinned calibration for `icv1-f2m13-t181-515ee569` (record `unpriced` is empty), S = 631 (492 to 806). Priced by measured instructions: S = (child Ir - 3.51 x 10^7) / 3265 / sqrt(2003) = 1.28 x 10^4. Ratio 20 against a proposed 1.5 tolerance: **control fails**, so every arm is priced in instructions here. Caveats: the calibration was pinned on an x86 Xeon, this is aarch64; the instruction figure contains Groebner work that is not word XORs; the 3265 was measured at n = 41. None explains a factor 20 by itself, and the control was not repeated on the same host as the calibration.
- The committed leaderboard K13 figure (inherited-F4 S = 16.29, `SPEC-K13`, 2026-10-11) is the unpriced lower bound; the same arm priced through the calibration in this tree is 631 (defect D3).

4. Root extraction (instrument built for this pilot)

msolve 0.9.5 at characteristic 2: plain `solve` runs its FGLM step but writes an **empty** output file, and `-P 1|2` writes nothing either (checked on a toy system). Roots are therefore read from the `-g 2` reduced basis: the standard monomials give the quotient algebra A (dimension = number of F_2 points; the ideal is radical because the field equations are in it), the multiplication matrices of the x-bits are built by normal forms and A is split into joint eigenspaces (an FGLM-type step, `msolve_s4.rs`). Its native work is counted (`extraction_native_ops`, at most 4.8 x 10^3 per target, negligible) and uncharged in S. Checked against brute force over V^3 (every ordered triple with `S_4 = 0`): **14/14** targets at n = 9 in each form (6 non-unit ideals, largest 46 points), **3/3** at n = 13 (`probes/`). The unit-ideal verdicts (no relation) agree with brute force too.

5. Cells

`S` = GAE per sqrt(r), one target, cold, priced in measured instructions at 3265 per GAE unless marked "uncharged".

cell	published-arm S	lab-arm S	ratio	verified	status
K_0 n=9, r=127, m=3, l=4, 8 targets	2.8e7 (1.9e7 to 4.2e7)	9.5e3 (8.0e3 to 1.2e4)	2.6e3 in instructions (2.2e3 to 3.1e3)	8/8	measured
(same cell), solver uncharged	36.0	36.0	tie (same query stream)		
(same cell) vs matched rho	4.8e6 times rho	1.6e3 times rho (rho S 5.90)			
K_0 n=13, r=2003, m=3, l=5	not run; projection 1.4e9 (e_i), 3.1e8 (x-only)	1.28e4 (instructions); 631 (native, pinned calibration); 16.29 (leaderboard, unpriced)	1.1e5 (e_i) and 2.4e4 (x-only), projection	one system checked	did not complete
other binary, Koblitz 41-61, m=4, prime n=4,5	not run				not run

Notes.

- **n = 9.** Instance `icv1-f2m9-t5-81e744be`. Published-arm S = (group GAE + msolve Ir / 3265) / sqrt(127) per target; lab S = (child Ir - 3.51 x 10^7) / 3265 / sqrt(127); median and range over the 8 targets. No pinned calibration exists for n = 9, so native-count pricing of the lab arm is not available there; `ecbench` leaves the solvers uncharged and both IC arms report S = 36.0. All three arms verified 8/8; the msolve arm repeated gave identical S.
- **AGENTS.md section 5a band** (unpriced native work at 0, 0.1, 1 GAE per op; lab ops = word XORs, msolve ops = instructions retired): S at 0: lab 36.0, msolve 36.0 (**tie**); at 0.1: 5.5e3 and 9.2e9; at 1: 5.5e4 and 9.2e10. The order, msolve above lab, holds at 0.1 and 1 and ties at 0; the instruction measurement (ratio 2.6e3) is the primary comparison. The other uncharged counters (`as_solves`, `row_ops`, `frobenius_maps`, `target_guard_probes`) are equal on both IC arms and rho has none that were not swept.
- **n = 13 projection.** 36 systems per target (median solver calls of the committed K13 inherited-F4 records) x Ir per system (5.70e12 e_i form, one system; 1.27e12 x-only, two systems) / 3265 / sqrt(2003), and the same product over the measured lab child Ir (1.91e9). An `ecbench` attempt at this cell was stopped at 5 min CPU and 4.5 GB RSS with the first system unfinished (`runs/C1_k13_aborted_process_time.txt`).

- **Not run.** Random binary: the symmetrised `S_4` is specialised to b = 1 (D6). Other sizes: the first system is already 1e12 to 1e13 Ir at n = 13.

Tuning probe (not a cell, per relation system, K_0): x-only form against `e_i` form, n = 9, l = 4 (14 targets): mean 1.21 x 10^10 against 6.49 x 10^10 Ir (5.4x); n = 13, l = 5: 1.27 x 10^12 against 5.70 x 10^12 (4.5x). Lab arm, whole run per solver call: 2.5 x 10^7 (n = 9), 5.3 x 10^7 (n = 13). The cheaper form is still 5 x 10^2 (n = 9) to 2 x 10^4 (n = 13) times the lab per system.

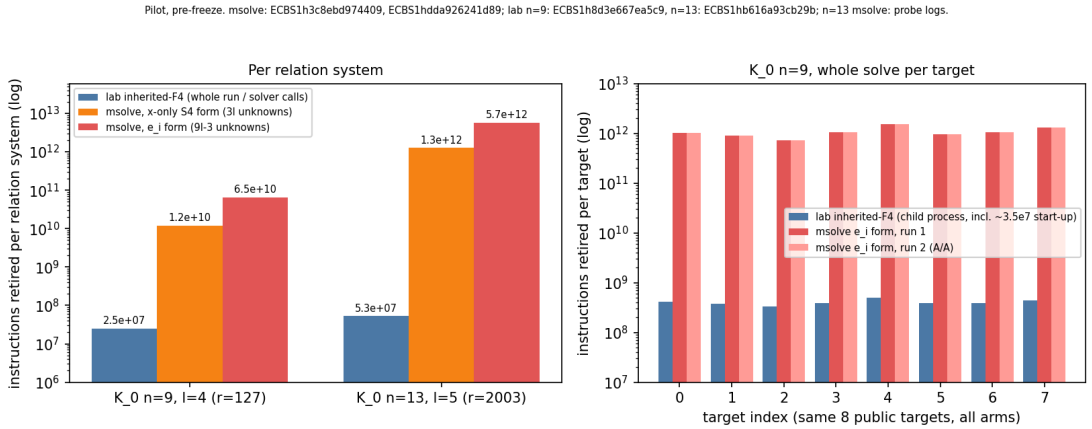


Figure 1: Per relation system and per target (recorded values).

6. Instrument defects found

- **D1. Worktrees share inodes.** `src/cryptanalysis/inherited_f4.rs` has 6 hard links across worktrees and `src/bin/ecbench/pdp3_koblitz.rs` 13. An in-place edit in one worktree appears in the others: this worktree showed another session's uncommitted `inherited_f4.rs` change (adds `mod profile;` with no `profile.rs`, so the tree would not build). *Effect:* an in-place edit here would have altered 12 other worktrees. I replaced my `pdp3_koblitz.rs` by a new inode before editing and **left `inherited_f4.rs` untouched and uncommitted** (not my change).
- **D2. A sparse worktree cannot build.** `Cargo.toml` names 40+ `[[bin]]`/`[[example]]` paths outside the sparse set, and `include_str!` reads `docs/` and `research/` files. Built from a `git archive` subset in a separate directory; cold build 5.4 min.
- **D3. `ecbench` S is blind to an unrecognised solver unit, and the leaderboard K13 figure is stale.** At n = 9 the msolve and lab arms carry the same S (36.0), and the framework's pre-removal figure treated msolve's wall nanoseconds as GAE (`framework_total_gae_before_solver_removal` 1.03e11). At K13 the lab arm is priced through a pinned calibration in this tree (S 631), against the leaderboard's 16.29 (unpriced, 2026-10-11): a 39-fold difference in the headline IC number for the same arm.
- **D4.** `ecbench` holds one global lock `/tmp/crypto-bench.lock`; a second session exits at once unless `--wait` is given. `isolation_required` must be `L0` here (`"none"` is rejected; `L1` cannot be earned).
- **D5.** msolve 0.9.5 returns no solutions at p = 2 (section 4). `instructions retired` needs the sandbox off (a sysctl). The `ecbench` parent's figure excludes **every** per-run child (all arms), and identical sessions differ in the parent by about 1 %; per-run figures need the shim.
- **D6. Protocol defects.** (a) PROTOCOL section 1 says the reference is `e_i`-symmetrised, section 2 says "T-symmetrised"; the code has no `T` invariants. (b) B3/B4 list random binary

curves, but `weil_descend_s4` asserts `b = 1`, so no random binary cell is runnable (needs M1). (c) Section 4′s Callgrind control cannot run on this host, and the instruction-based substitute failed the proposed tolerance (section 3).

- **D7.** The conductor control plane at 127.0.0.1:8491 refused connections; no scope claim was made (hooks let edits through). `gh` and AWS credentials were not used; the checkpoint token worked.

7. What is needed to go further

- **WDSat**: `github.com/mtrimoska/WDSat` at `61c6ff3f49445af2729274819edb27b85a89efc1` (GPL-3.0, external process, build from `src/` with `make`, per-cell `config.h` constants); not fetched, size not checked, absent from this host. Needs an instance emitter (M3).
- **A Linux host with Valgrind** for the protocol's Callgrind `Ir`, and to repeat the cross-unit control on the host that pinned the calibration.
- **msolve 0.6.5 built from source** (`research/dreg_closure_m4l6_20261018/pod/boot.sh`) for the version A/A against 0.9.5.
- **M1**: general-b `S_4` (random binary), the `T`-invariants, and `m = 4`; **tuning sweeps** (specialisation/hybrid, `l`, `LP`) so the reference gets the tuning parity of PROTOCOL section 6. Without them the n >= 13 cells are not affordable (about 2 x 10^14 Ir per target projected, e_i form).
- The `benchmarks/<date>-<slug>/` copy of the graphs and PDF is deferred to whoever opens a PR (that path is outside this sparse checkout).

8. Files

All under `pilot_20261030/`: `specs/` (the specs run), `runs/` (records, session, host and process time of every session; `*.tm` are the per-child `time -l` files of the shim runs; the aborted K13 attempt's process time), `probes/` (extraction check, n = 13 systems), `graphs/` (SVG, PDF, 150 dpi PNG, data), `ecbench_child_time_shim.patch`. Code: `src/bin/ecbench/msolve_s4.rs`, `examples/msolve_s4_probe.rs`, and the `msolve-s4` engine in `src/bin/ecbench/pdp3_koblitz.rs`. Derived values in section 5 are arithmetic on the recorded values, shown in the notes; they were computed with a throwaway script, not committed as research code.